Reproducing “Uncertainty-Aware PCA Revisited”

- Friesecke, Braune, Rössl, Theisel – IEEE VIS 2025
- Problem: classic PCA ignores uncertainty in data points; prior uncertainty-aware PCA (Görtler et al. 2019) ignored the uncertainty of the global covariance itself
- Contributions: (1) covariance stability glyph – shows how stable the top-3 eigenvectors are; (2) sampling-based projection – run PCA on U random realizations and draw density contours
- Code: github.com/lfriesecke/uncertainty-aware-pca-revisited

Setup & what broke

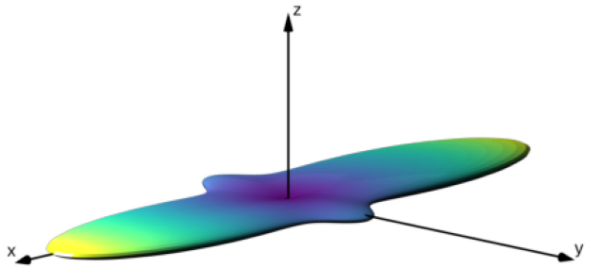
- README: python -m venv venv_uapcar → pip install -r requirements.txt → python example_*.py
- Error: numpy==2.3.1 requires Python ≥ 3.11 (I have 3.10.10) → pinned numpy 2.2.6, no code changes needed
- Runtime: U = 1,000,000 ≈ 2 h per dataset on my laptop → used U = 100,000 (paper's Fig. 10 shows this is close)
- Glyph code only exports an .OFF mesh → wrote my own matplotlib 3D renderer
- Glyph axes didn't match the paper → found commented-out "idx = [2,1,0] # TODO: ask if wanted" in glyph.py; applying it (x↔z swap) reproduces the paper exactly

Fig. 5 – IRIS glyph

- Long flat lobes and only small bumps: the eigenvectors are fairly unstable in the xy-plane
- Shape matches the original; only the shading differs

Fig. 5 (left) – IRIS covariance stability glyph

Original (paper PDF)



Reproduced (glyph_iris_swapped)

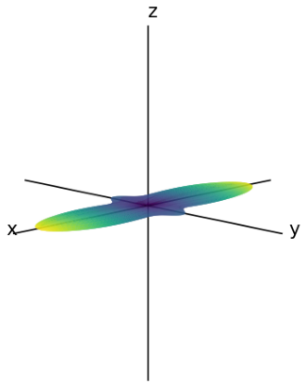
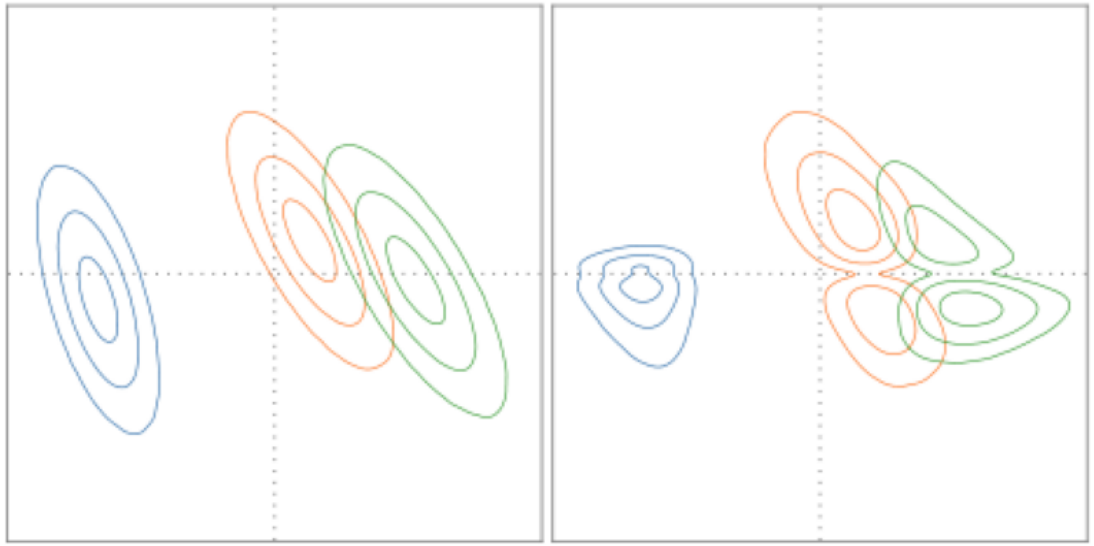


Fig. 5 – IRIS projection

Fig. 5 (right) – IRIS sampling-based projection

Original (paper PDF)



Reproduced (projection_iris_U100000)

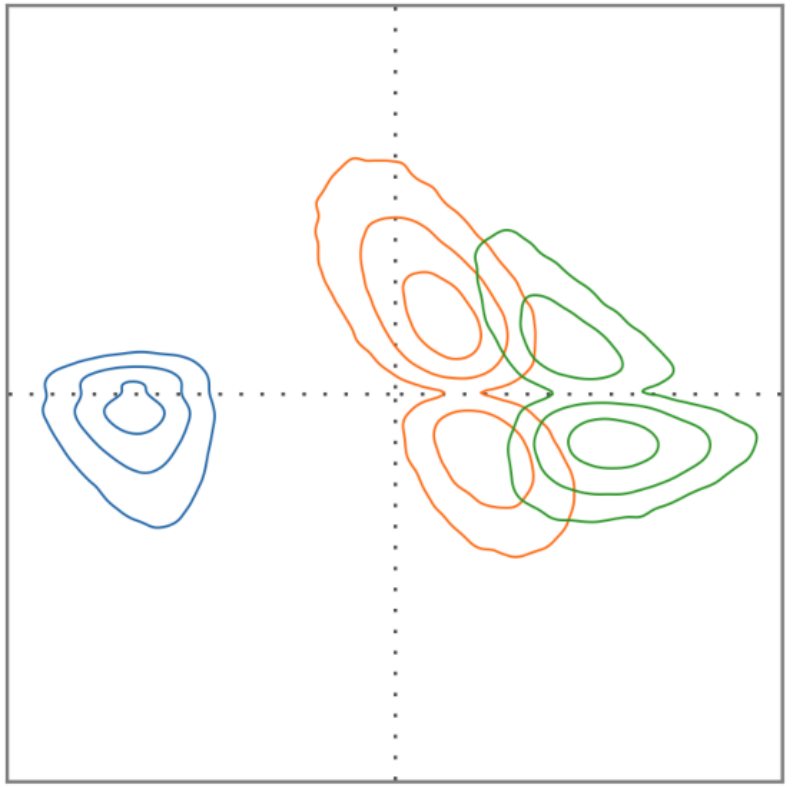
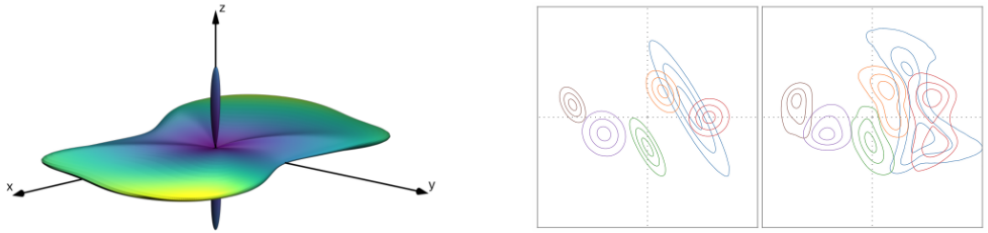


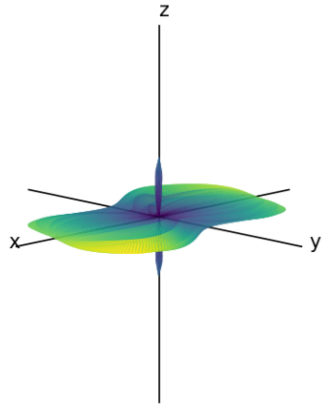
Fig. 8 – STUDENT glyph + projection

Fig. 8 – STUDENT glyph (left) and projection (right)

Original (paper PDF)



Reproduced (glyph_students_swapped)



Reproduced (projection_students_U100000)

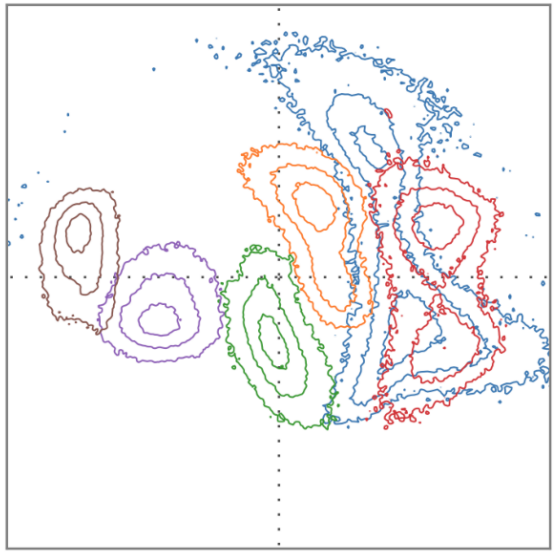
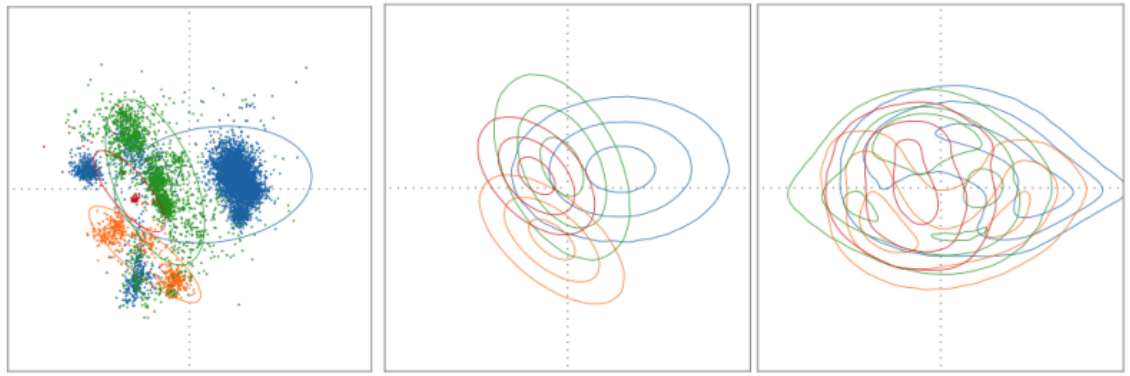


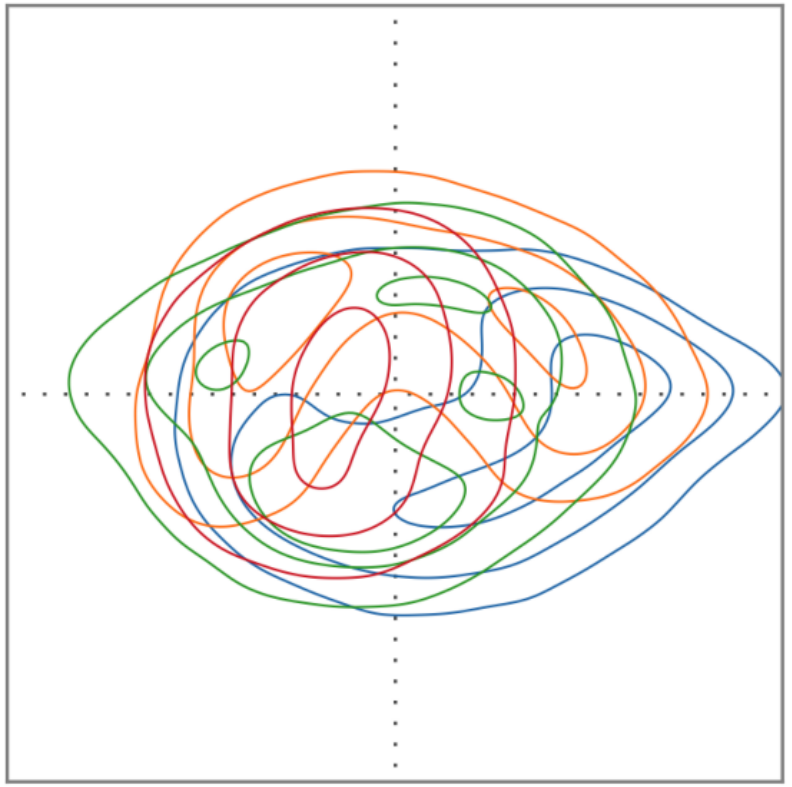
Fig. 9 – ANURAN CALLS projection

Fig. 9 (right) – ANURAN CALLS sampling-based projection

Original (paper PDF)



Reproduced (projection_anuran_U100000)



Takeaways

- All 4 visualizations reproduced: same structure, clusters, orientation
- Differences: noisier contours (U=100k vs 1M), different 3D renderer, x/z axis order in released glyph code
- Lesson: pinned requirements can block older Pythons; check for leftover TODOs when results look “rotated”
- Reproducibility score: high – public code + bundled datasets + fixed random seeds